

QCM

1. → ① → ①

2. → C

3. → ①

4. → C

5. → A

6. → B

7. → A

8. → C

9. → D

10. → B

11. → B

12. → C

13. → B

14. → C

15. → B

16. → C

17. → A

18. → C

19. → A

20. → B

Answer all the questions:

Q1. Definition of supervised learning is a strategy to predict the data by given dataset. This learning from the data or a lot of data in order to categorise of data or classify of data to the correct type.

Q2. M1 give 99% of accuracy for training dataset and gives 30% of accuracy for testing dataset. It means that machine learning M1:

- Supervised machine learning
- Training more than testing or predicting
- More accuracy of predicting the data in the future
- Better machine learning for analysing, controlling data and keep the data

Q3. Regression Analysis is the analysis between dependent variable to predict the independent variable. according to the relationship from both of them and dataset. Learn from dependent or accurate data to predict the independent or inaccurate data.

Q4. Elucidate any two challenges of Machine Learning:

- Scalability
- Capacity
- Speed
- Redundancy

25. List out any two major applications of data science:

- Sell Order Prediction : Help business of selling product.
- Video or audio Prediction: Train the module how to predict the given data by learning from a lots of data.

26. Distinguish between the terms of Structure and Semi-Structured data
+ Structured Data

- Defined data
- Row and Column (Table)
- Categorised data
- Cleared datatype and dataset
- Easy to understand and learning

+ Semi-Structured Data

- Uncompletely data set
- Half structured and half un-structured
- Most of data are structured
- Better for learning and testing
- Need more improvement.

27. Find out an Outlier in the Data:

- Used outlier module to see the data
- Boxplot, Seaborn, Linear Regression module...
- Define minimum and maximum scope of data in order to see
- Formula: $\text{Min}(Q - 1.5IQR)$ & $\text{Max}(Q + 1.5IQR)$.

28. Data Pre-processing is the type of data which already defined or classify into type of data and it is quality data for good predicting and get a high accuracy result.

29. Data normalization is required in K-NN (K Nearest Neighbor) because:

- Easy to categorise the data
- High accuracy
- Small or less distortion
- Data already clean and good for making the decision.
- Speed faster.

30. Continuous attributes are discretized: It is the natural of data that we can not change. We can not represent as numerical data that is the natural data. But is true that continuous attributes are discretized.

31. K in K-means algorithm is the represent number of group or type of data number.

32. Name few hyperparameters of machine learning algorithm:

- Binary
- Multinomial
- Radio.

33. Gradient Descent it is the unsupervised learning for prediction the data.

34. Based on the given linear boundaries of SVM

- The third 3rd picture is greater than 1st and 2nd
- Long distance and short distance between line
- Short is better but everything is better based on the decision

35. Radial Basis Function Kernel work in SVM:

- The function will transform the dimension d into multiple dimension in polynomial.